# Sequential Circuits - The JK Flip-Flop

## 1. SR vs. JK Behavior

The primary purpose of the JK Latch is to resolve the undefined/invalid state of the SR Latch. By crisscrossing the final outputs back to the input gates, the circuit uses its own current state to determine how to handle a `1, 1` input.

As shown , the function tables compare directly:

### SR Latch Function:

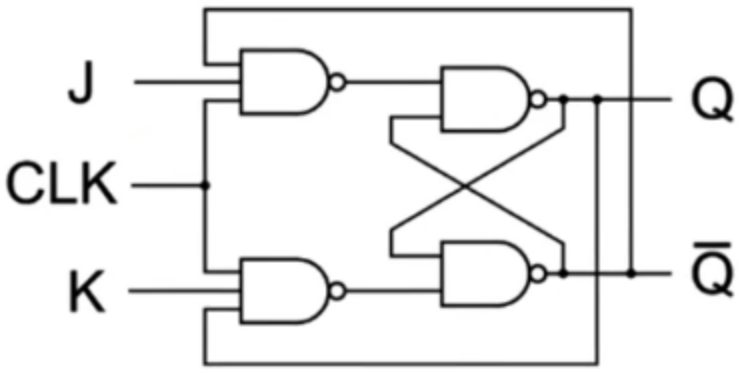
| $S$ | $R$ | $Q$ |
|---|---|---|
| 0 | 0 | Hold |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | Invalid |

### JK Latch Function:

| $J$ | $K$ | $Q_n$ |
|---|---|---|
| 0 | 0 | Hold |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | Toggle |

## 2. Circuit Diagram (NAND Implementation)

This is the schematic you drew at the top of the page. Notice how it is essentially a clocked SR latch, but with the outputs $Q$ and $\overline{Q}$ fed all the way back into the front steering NAND gates.



## 3. The Analytical Tables

These are the tables which explicitly map out the math behind the JK logic.

### A. Characteristic Table

This table defines exactly what the next state ($Q_{n+1}$) will be based on the current inputs ( $J$, $K$) and the current state ($Q_n$).

| $J$ | $K$ | $Q_n$ | $Q_{n+1}$ |
|---|---|---|---|
| $0$ | $0$ | $0$ | $0$ |
| $0$ | $0$ | $1$ | $1$ |
| $0$ | $1$ | $0$ | $0$ |
| $0$ | $1$ | $1$ | $0$ |
| $1$ | $0$ | $0$ | $1$ |
| $1$ | $0$ | $1$ | $1$ |
| $1$ | $1$ | $0$ | $1$ |
| $1$ | $1$ | $1$ | $0$ |

*(Notice the last two rows: When $J = 1$ and $K = 1$, the next state is always the exact opposite of the current state $Q_n$. This is the Toggle feature in action).*

### B. Excitation Table

This table is used backward for circuit design. It tells you what inputs to apply to $J$ and $K$ if you want to force a specific state transition.

| $Q_n$ | $Q_{n+1}$ | $J$ | $K$ |
|---|---|---|---|
| $0$ | $0$ | $0$ | $X$ |
| $0$ | $1$ | $1$ | $X$ |
| $1$ | $0$ | $X$ | $1$ |
| $1$ | $1$ | $X$ | $0$ |

## 4. K-Map Derivation for the Characteristic Equation

To find the equation for the Next State ($Q_{n+1}$), we plot the outputs from the JK Characteristic Table onto a 3-variable K-Map based on the inputs $J$, $K$, and $Q_n$.

Unlike the SR Flip-Flop, the JK Flip-Flop has no "Invalid" states, meaning there are absolutely no "Don't Cares" ($X$) to use as wildcards. We must rely entirely on the definitive `1` s.

| $J\backslash KQ_n$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 1 |

### Grouping the K-Map:

We have four `1`s on the board, and we can group them into two distinct pairs (loops of 2).

1. **Group 1 (The Bottom Corners):** Look at the bottom row ($J = 1$). There is a `1` in the `00` column and a `1` in the `10` column. Because K-Maps wrap around the edges, we can group these two corners together.
   - Across these two cells, $K$ changes from $0$ to $1$, so it is eliminated.
   - The constant variables are $J = 1$ and $Q_n = 0$.
   - **Resulting Term:** $J\overline{Q}_n$
2. **Group 2 (The Vertical Column):** Look at the column where $K = 0$ and $Q_n = 1$ (the `01` column). There is a `1` in the top row and a `1` in the bottom row. We group these vertically.
   - Across these two cells, $J$ changes from $0$ to $1$, so it is eliminated.
   - The constant variables are $K = 0$ and $Q_n = 1$.
   - **Resulting Term:** $\overline{K}Q_n$

### The Final Derived Equation:

By ORing these two groups together, we get the minimized Characteristic Equation that governs the JK Flip-Flop:

$$Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n$$